Article

MA-SQLGrid: An Auditable Coordination and Evaluation Framework for Power-Grid Text-to-SQL

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Featured Application

MA-SQLGrid is a research framework for traceable natural-language access to structured power-grid records. It exposes intent, schema, candidate, execution, and selection evidence so that generated SQL and returned rows can be inspected before consequential use.

Abstract

Power-grid databases encode asset, maintenance, temporal, and topological semantics, so executable SQL can answer the wrong question. We present MA-SQLGrid, an auditable five-role architecture separating intent analysis, schema mapping, candidate normalization, read-only execution validation, and constructed-state evidence through a shared blackboard and deterministic controller. Robustness here denotes tested mutation denial, bounded execution, and fail-closed handling of incomplete evidence. Evaluation covers GridDB generation, component, and constructed-state studies; BIRD Mini-Dev across 11 public databases; and a frozen historical pool of eight candidates for each of 180 GridDB questions. Context and presented-value effects varied with the frozen model snapshot. A unified shape-and-denotation evaluator scored C000 at 76/180, validation-aware ranking at 99/180, and complete three-witness selection at 100/180; the best fixed source, Qwen F01, reached 129/180. Evidence-aware selectors frequently tied at the highest score and were sensitive to candidate order, indicating that the available evidence often could not distinguish semantically competing candidates. MA-SQLGrid provides a traceable interface for power-grid Text-to-SQL evaluation, but the experiments do not establish an end-to-end accuracy advantage from five-role coordination or superiority over the best fixed source. Broader efficacy requires unseen, expert-reviewed grid queries and budget-matched generation.

Keywords: text-to-SQL; multi-agent systems; power-grid databases; schema grounding; execution validation; constructed-state testing; reproducibility

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1. Introduction

Power-grid organizations maintain relational data about assets, locations, topology, sensor readings, work orders, technicians, and maintenance activity. SQL provides precise access to these records, but it requires knowledge of table names, join keys, identifiers, units, aggregation conventions, and local coding rules. Text-to-SQL systems aim to reduce this access barrier by translating a natural-language request into an executable query. In an engineering workflow, however, a fluent query is not sufficient evidence of correctness. A query can execute while selecting the wrong status field, omitting a required filter, using an unintended join path, or returning the expected number of columns with the wrong content.

The risk is amplified by the semantics of grid records. An identifier that looks numeric may be a categorical code; a timestamp can refer to sensing, issuance, acknowledgement, or closure; and an asset status may describe the asset, an event, or a work order. The same request may admit several executable queries with different engineering meanings. An auditable workflow should expose how the request was interpreted, how schema and values were grounded, what SQL candidates were considered, which candidates executed, and why one candidate was selected. When the available evidence is incomplete, the workflow should be able to abstain instead of concealing the ambiguity.

Cross-database benchmarks such as Spider and BIRD have made schema generalization and realistic database content central evaluation concerns [1,2]. Spider 2.0 extends this setting to enterprise workflows with large schemas and multiple SQL dialects, while Multi-SQL broadens context-dependent evaluation across diverse operations [3,4]. Schema-aware models, constrained decoding, context selection, decomposition, and execution guidance address complementary failure modes [5–9]. Recent systems also use multiple roles for generation, review, or repair [10–13]. Most studies improve one or more of these stages, but fewer treat generation, database-enforced validation, state evidence, candidate selection, and trace retention as a single reproducible decision process. This gap matters for grid data because successful execution provides only weak evidence about field meaning, time scope, status coding, or join intent.

MA-SQLGrid addresses this systems problem through five bounded role interfaces coordinated as a multi-agent architecture. The Query Analyst represents question intent; the Schema Cartographer maps that intent to tables, columns, values, and relationships; the SQL Synthesizer normalizes and identifies candidate statements; the Validation Engine records read-only execution evidence; and the Metamorphic-State Critic records outcomes on named constructed states. A deterministic controller coordinates these roles through a shared blackboard trace and either selects an eligible candidate or abstains. The controller is not a sixth reasoning agent; it enforces message order, evidence requirements, and the registered selection rule. This separation makes errors localizable: a reviewer can determine whether a failure arose during intent analysis, schema mapping, candidate ingestion or normalization, execution, state testing, or adjudication.

The empirical design examines this workflow from complementary angles. GridDB experiments vary schema context, structural hints, and presented values on a controlled power-grid schema. A constructed-state study examines whether observed results persist across registered database transformations. BIRD Mini-Dev tests the portability of frozen prompting procedures across 11 public, non-grid databases. Finally, a no-generation study applies fixed-order and evidence-aware selectors to the same historical pool of eight candidates per GridDB question. The studies are not combined into a single accuracy benchmark. Each instead identifies where context, validation evidence, or deterministic selection affects observable behavior.

The study addresses three research questions:

1. How do schema context, value evidence, and prompting organization affect executable results for frozen models on GridDB and public databases?
2. When candidate generation is held fixed in a historical pool, how does evidence-aware adjudication change the selected query and its reference match relative to fixed order?
3. How stable are these observations across model snapshots, constructed database states, score ties, and candidate order?

The paper makes three contributions. First, it defines MA-SQLGrid, a five-role framework that integrates intent analysis, schema mapping, candidate normalization, read-only execution validation, and state evidence in a traceable blackboard workflow with explicit boundaries between model-backed and deterministic components. Second, it provides an evaluation design spanning GridDB, BIRD, constructed states, and a fixed historical candidate pool, allowing generation behavior, portability, validation behavior, and candidate selection to be observed separately. Third, under one shape-and-denotation evaluator, it reports 76 C000, 99 validation-aware, and 100 complete-witness matches in the historical pool, alongside all eight fixed-source results; the best fixed source reached 129, while high tie rates and candidate-order sensitivity identify a remaining semantic-ranking bottleneck.

In this paper, *robustness under tested conditions* refers to mutation denial, bounded read-only execution, complete-evidence gating, deterministic failure handling, and the observed response to candidate-order perturbation. The term *multi-agent* denotes five specialized role interfaces and their coordinated trace; some roles are deterministic, and the controller is not a sixth reasoning agent. The present experiments do not estimate a causal five-role advantage over a call-matched single agent, and the historical-pool analysis remains descriptive. Section 5 interprets the empirical findings within these scope conditions.

2. Related Work

2.1. Schema Linking, Context Construction, and Evaluation

Text-to-SQL systems map a natural-language request and database schema to an executable query. Spider established cross-domain evaluation on unseen databases, and BIRD expanded the setting toward larger databases with richer content [1,2]. Spider 2.0 and MultiSQL further emphasize enterprise-scale schemas, varied SQL operations, and context-dependent workflows [3,4]. These benchmarks make schema generalization visible, but their scores still depend on the database engine, evaluator, split, and model configuration.

Schema linking connects question spans to tables, columns, values, and foreign-key relations [5,14,15]. Full-schema context protects recall but increases token cost and distraction. Compact context reduces the search space but can omit a bridge table or coded field. Retrieval, graph pathfinding, and bidirectional table–column selection seek a better balance [16–21]. Value examples add another signal because the intended field may be clearer from stored codes than from its name. The GridDB studies in this paper examine these context and value choices as observable components of a larger workflow.

Evaluation also needs more than one notion of match. String comparison can penalize equivalent SQL, while result equality can reward different queries that happen to coincide on one database snapshot. Distilled test suites evaluate denotation across multiple database states to expose some accidental equivalence [22]. MA-SQLGrid separately reports strict execution equality, constructed-state agreement, and bounded metamorphic invariance. These endpoints describe execution behavior under the stated transformations; expert-reviewed business semantics remain a separate validation layer.

2.2. Multi-Role Generation, Validation, and Selection

Constrained decoding, decomposition, contextual grounding, and execution-guided repair address different stages of SQL production [6–9,23]. Recent workflows add question rewriting, attention-oriented decomposition, partition-and-drill reasoning, preference optimization, active database probing, or dialect-aware bootstrapping [24–29]. These methods mainly improve generation or repair. MA-SQLGrid focuses on the interfaces that connect candidate production to database-enforced validation and final selection.

SQL-oriented language-model teams distinguish candidate diversity from explicit role coordination [10]. Multi-agent Text-to-SQL systems have used decomposition and tools, question-dependent routing, and review–rebuttal–revision consensus [11–13]. MA-SQLGrid uses *multi-agent* to denote five stable role interfaces recorded in a shared blackboard trace. The SQL Synthesizer may package externally produced candidates, whereas validation, state analysis, and final selection use explicit records and deterministic rules. Reproducibility therefore applies to these handoffs and the adjudication procedure, not to multi-model dialogue.

Read-only execution and evidence-aware adjudication are complementary. Database controls prevent an admissible evaluation from mutating the source, whereas adjudication compares only candidates that satisfy the required safety, execution, and state-evidence gates. A stored trace cannot guarantee sound reasoning, but it can reveal whether an error entered through intent analysis, schema mapping, generation, execution, or selection. This diagnostic role motivates the framework even when candidate generation is held fixed.

2.3. Power-Grid Text-to-SQL

Power-system data combine topology, equipment attributes, measurements, time series, costs, constraints, and maintenance records. RTS-GMLC and SimBench provide reproducible engineering structures that can be transformed into relational databases [30,31]. Their availability does not by itself produce an expert-reviewed Text-to-SQL benchmark: intended projections, units, ordering, tie handling, and time semantics still require domain adjudication.

DKA-SQL is a direct power-grid Text-to-SQL precedent that adapts domain knowledge to grid questions [32]. Its corpus and evaluation boundary differ from the present study. MA-SQLGrid instead investigates a systems architecture for traceable role coordination, read-only validation, constructed-state evidence, and deterministic candidate selection. GridDB supplies a controlled grid case study, BIRD supplies public cross-database portability evidence, and RTS-GMLC/SimBench assets identify a path toward broader structural validation. Table 1 summarizes this positioning without treating results from different datasets and call budgets as directly comparable.

Table 1. Positioning against adjacent Text-to-SQL approaches; entries describe reported design emphasis, not comparable performance.

Approach	Principal emphasis	Evaluation setting	Difference from MA-SQLGrid
RGISQL [33]	Grammatical and relational graph modeling	Public Text-to-SQL tasks	Learned generator rather than a traced executor/adjudicator boundary
Zero-shot prompting [34]	Prompt strategies and LLM behavior	Public benchmarks	Prompt comparison without the five-role blackboard
Schema retrieval [16]	Embedding/vector-store context selection	LLM SQL generation	Retrieval quality rather than fail-closed named-state selection
DKA-SQL [32]	Domain knowledge adaptation	Power-grid Text-to-SQL	Complementary domain-adaptation focus; MA-SQLGrid studies traceable validation and candidate selection
MA-SQLGrid	Five-role trace, read-only validation, state evidence, and deterministic selection	GridDB case study plus BIRD portability	Integrates generation-to-selection evidence for power-grid Text-to-SQL

3. Materials and Methods

3.1. Problem Formulation and System Overview

For question x_i, read-only database D_i, and introspected schema S_i, a generator produces one or more candidates

$$\mathcal{Y}_i = \{\widetilde{y}_{i1}, \ldots, \widetilde{y}_{ik}\}. \tag{1}$$

Each candidate must satisfy a single-statement read-only contract. The accepted leading form is `SELECT` or `WITH`; comments, multiple statements, write operations, schema changes, attachments, and pragmas are rejected. A candidate is eligible for adjudication only when it is safe and executable.

The offline evaluator may later compare the selected query with a reference result. Let $E_i = 1$ when the selected SQL executes and its result equals the frozen reference result under the registered evaluator, and $E_i = 0$ otherwise. Reference SQL, reference results, E_i, and any derivative correctness field are excluded from the inputs to all five roles and the deterministic controller. Evaluation occurs only after the blackboard is sealed.

This boundary yields three distinct states. A candidate is *admissible* when its syntax and requested operations satisfy policy, *executable* when it completes within the registered database limits, and *correct under the frozen evaluator* when its result matches the reference after sealing. Admissibility is neither executability nor correctness. The manuscript preserves all three labels because collapsing them would let an executable but semantically wrong query appear validated.

The framework's response contract contains the selected SQL, stable candidate and source identifiers, referenced tables and columns, execution-state and result hashes, state-coverage summary, abstention or selection reason, and the sealed trace digest. Figure 1 shows how these records move through the five-role architecture. The controller is not a sixth reasoning model: it schedules the roles, checks their typed messages, applies the deterministic rule, and seals the shared blackboard trace before offline evaluation.

The response contract supports traceability, not user authorization. A calling application must authenticate the user, restrict the accessible database view, and decide whether returned rows may be displayed. Generated SQL and execution success do not replace those institutional controls.

3.2. Datasets and Evaluation Roles

GridDB-Maintenance-v2 v0.1 is a synthetic SQLite case study with eight tables and 98 rows: `asset_types` (6), `assets` (18), `grid_topology` (9), `locations` (8), `maintenance_logs` (8), `sensor_readings` (26), `technicians` (8), and `work_orders` (15). It contains 200 natural-language–SQL records: 20 development records and 180 factorial evaluation records. The evaluation records include 66 easy, 91 medium, and 23 hard items and map to 70 normalized gold-SQL structural clusters, of which 58 are singletons. The evaluation partition had been visible during project development; it is a controlled finite-set case study, not a sealed production benchmark.

BIRD Mini-Dev contributes 500 public items over 11 SQLite databases. The protocol fixed item order, prompts, two local models, evaluator, Python 3.10.11/SQLite 3.40.1, and zero retries. Each backbone made 2500 calls and produced 2000 final predictions; independent re-execution found zero score mismatches. Two incomplete predecessor runs were excluded from every scientific denominator and retained only in the supplementary execution record.

The GridDB reliability release contains 18 states: the canonical snapshot, three insertion permutations, and 14 additional schema-valid stress states. The scorer produced 25,920

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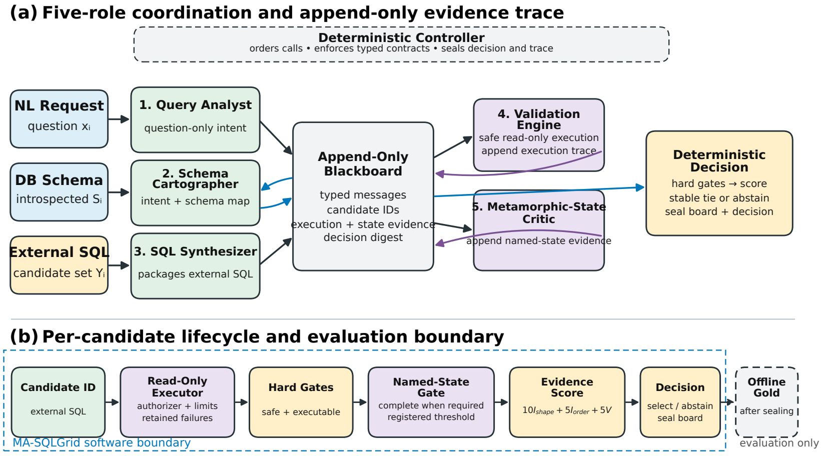

Figure 1. MA-SQLGrid information flow and candidate lifecycle. **(a)** The Query Analyst and Schema Cartographer post question and schema records to the shared blackboard, and the SQL Synthesizer normalizes externally supplied candidates. The Validation Engine records read-only execution evidence, while the Metamorphic-State Critic records named-state evidence when required. **(b)** Hard eligibility gates precede the effective evidence score $e(y) = 10I_{\text{shape}} + 5I_{\text{order}} + 5V$, $V \in [0, 1]$, and the deterministic tie or abstention rule. Reference results are loaded only after the decision and trace digest are sealed.

prediction-state rows (1440 × 18). A prediction-blinded automated protocol admitted a 66-question subset for the primary 15-state analysis and retained 114 order-sensitive questions as diagnostics. These constructed states are not operator-certified grid snapshots.

The evaluation distinguishes inherited predictions, deterministic recomputations, new protocol-bound runs, and diagnostic implementation records. These labels preserve chronology without treating software replay as an accuracy experiment. Table 2 summarizes the three resources that enter the main empirical argument. RTS-GMLC and SimBench remain potential external-validation resources because their current question–SQL pairs have not completed qualified-human semantic review.

Table 2. Resources used in the main empirical argument. Each resource answers a different research question and retains its own denominator.

Resource	Role in the study	Sample or unit	Primary endpoint
GridDB v0.1	Controlled grid-schema generation, component, and historical-pool studies	1 database; 8 tables; 98 rows; 20 development + 180 evaluation questions	Strict execution equality, component contrasts, and reference match after selection
Constructed-state suite	Stability of retained predictions under registered database transformations	18 states; 1440 predictions; 66-question primary subset	Fifteen-state logical-AND agreement and witness coverage
BIRD Mini-Dev	Portability of the prompting and execution workflow to public non-grid databases	500 items across 11 SQLite databases	Execution accuracy and database-grouped method contrasts

3.3. Experimental Protocols and Evaluation Boundaries

Table 3 maps each protocol to its unit, endpoint, and evidence boundary. The studies answer complementary questions and are interpreted separately because their candidate sources, call budgets, and visibility histories differ.

Table 3. Experimental design and inferential role of each protocol. Composition intervals describe sensitivity to the observed finite-corpus grouping, not population confidence.

Study	Unit/groups	Primary endpoint	Research role
GridDB factorial	180 questions/cell; 70 normalized-SQL groups	Execution equality with nine-test Holm correction	Context and hint effects for two model snapshots
Component E1/E2	170 paired or 180 questions; 61 or 70 groups	Execution equality and grouped composition intervals	Presented-value and multi-candidate-selection effects
Constructed states	66 questions; 12 structural groups	Fifteen-state logical AND; exact 2^{12} signs	Agreement under prediction-blind transformations
BIRD	500 items; 11 databases	Execution accuracy and 12 Holm-adjusted contrasts	Public non-grid portability of the workflow
Coordination replay	180 questions	Pool coverage and complete finite-pool counts	Trace and handoff reproducibility
Historical-pool selection	180 questions; shared 5760-attempt ledger	Reference match, rescue/harm, ties, and order sensitivity	Evidence-aware selection within the observed pool

GridDB factorial outcomes are inherited and development-visible; the component procedure was fixed before its calls on development-visible items; constructed states were prepared without reading predictions; and BIRD used a fixed public protocol. The historical-pool study sealed every decision before loading raw reference outcomes, but same-item derived outcomes had been accessed earlier in development. Its selector comparisons are consequently descriptive, although their execution is deterministic and traceable.

3.4. Five-Role Architecture and Coordination

MA-SQLGrid comprises five specialist role interfaces coordinated by the controller shown in Figure 1. Messages receive consecutive sequence numbers, and the complete trace is serialized canonically to a SHA-256 digest. Each role has restricted inputs and produces one typed output for the next stage.

The **Query Analyst** produces a structured question-only intent record containing lexical tokens, detected aggregations, ordering requirements, and an explicit limit when present. It does not select SQL or see correctness. The **Schema Cartographer** maps lexical intent to tables, columns, and foreign-key edges. The current skeleton uses deterministic overlap and records unmatched tokens; a richer GridDB adapter adds disclosed domain rules and graph expansion.

The **SQL Synthesizer** packages external SQL without model calls, canonicalizes formatting, removes exact duplicates while preserving first occurrence, and assigns stable identifiers. The **Validation Engine** used by the historical-pool study opens the database with `mode=ro&immutable=1`, enables `query_only`, disables extensions, and installs an authorizer. It denies writes, DDL, transactions, attachments, pragmas, metadata enumeration, dangerous functions, and reads outside table/column allowlists. Progress and fetch handlers enforce time, opcode, and row limits. Every attempt retains database/SQL hashes, failures, timing, row count, and result hash without hidden retry.

A later executor revision adds function allowlisting, projected-width, raw-cell-byte, and total-result-byte limits. Fourteen tests cover BLOB denial, total-budget and boundary behavior, trace retention, oversized text or results, and wide projections. These controls postdate the frozen historical candidate and evidence ledgers; they therefore strengthen the implementation boundary without changing those stored artifacts. Table and column allowlists constrain query scope but do not authenticate users or provide row-level authorization; both executor variants establish tested SQLite mutation denial and scoped execution bounds.

The threat model assumes untrusted SQL but a trusted host, SQLite library, configured allowlist, and immutable database. It does not address a compromised process, OS escape, side channels, built-in malicious extensions, or upstream authorization errors. Production use requires process isolation, least-privilege views, authenticated context, logging, and independent security review.

Failure handling is intentionally fail-closed and does not retry. A parse denial, authorizer denial, timeout, opcode limit, row limit, width limit, cell limit, or total-result limit creates an attempt record and makes that candidate ineligible for the current decision. The controller cannot silently weaken a limit or substitute a repaired query. A repair procedure may be studied as a separate registered generation step, as in BIRD B3, but its extra physical call and visibility must remain explicit.

The **Metamorphic-State Critic** (historically named the Counterfactual Critic) consumes named state results, rejects duplicate state identifiers, and records evaluated, passed, failed, and missing states. When state evidence is optional, missing evidence remains diagnostic and does not alter the previously defined no-state behavior. When the caller requires state evidence, coverage must equal the complete registered state set and the pass threshold must be met; otherwise the candidate is ineligible. An incomplete $1/1$ record therefore cannot outrank a complete $10/11$ record. Gold-relative labels from the constructed-state study remain forbidden during selection.

The blackboard is append-only at the application layer: a role posts a typed message and cannot rewrite an earlier message. The controller checks role, message type, sequence number, and required fields before acceptance. Canonical serialization fixes key order and whitespace before hashing. This makes two runs comparable at the trace level and exposes an unexpected hidden field or changed handoff. It does not prevent a malicious host process from altering memory or files; process isolation and remote attestation are outside scope.

Information flow is intentionally asymmetric. The Analyst receives the question but no candidate SQL. The Cartographer receives the question, intent record, and introspected schema. The Synthesizer receives external candidate strings and assigns identities but cannot execute them. The Validator receives one candidate plus a scoped executor. The Critic receives named-state outcomes but no reference correctness. Only the deterministic controller receives the eligible evidence summaries. Gold enters a separate offline evaluator after the decision message and board digest exist.

3.5. Read-Only Validation and Evidence-Aware Adjudication

Eligibility always requires safety and successful execution. The archived 100-point representation assigns 40 points for each of those two properties, followed by 10 for question-derived aggregate-shape conformity, 5 for question-derived ordering conformity, and up to 5 for lexical value hits. Because every ranked candidate has already passed both hard gates, the first 80 points are constant and contain no ranking information. The effective evidence score is therefore $e(y) = 10I_{\text{shape}}(y) + 5I_{\text{order}}(y) + 5V(y)$ for an eligible candidate y, with $V(y) \in [0,1]$. When named-state evidence is optional, the controller selects an $\arg\max e(y)$ and ties preserve the predefined candidate order. When named-state evidence is required, incomplete coverage or failure of the specified pass threshold makes y ineligible before the same ranking. If the eligible set is empty, the controller abstains. This gate–score–tie function is mathematically equivalent to the archived implementation and shows that safety and execution are eligibility controls rather than discriminative evidence. The reported weight, pass-threshold, and order analyses are descriptive because the project history does not establish that every design choice preceded access to derived outcomes for the 180 items.

The 40/40/10/5/5 allocation, equivalently hard safety and execution gates followed by a 10/5/5 evidence score, is an illustrative deterministic policy rather than a learned or calibrated weighting. Its formal invariants are that unsafe or non-executable candidates are never selected; required named-state coverage is complete before threshold evaluation; gold-derived correctness is absent from the decision record; equal scores use the predefined order; and an empty eligible set yields abstention. These invariants support conformance testing and replay determinism, while semantic correctness and calibrated confidence require additional evidence.

Question-derived shape checks are deliberately weak. They detect cues such as "how many", "highest", "first", and explicit limits and compare them with the projected aggregate or ordering structure. They do not infer business semantics such as whether "active" refers to an asset, alarm, or work-order state. Lexical value hits likewise reward only presented values traceable to the question and context; they are not evidence that the associated column is the intended one. These low-bandwidth features keep adjudication inspectable but explain why large tie sets remain possible.

Abstention has two layers. Pre-ranking abstention occurs when no candidate passes safety and execution requirements or when required state coverage is incomplete. Post-ranking ambiguity is currently resolved by stable order because no pre-registered margin threshold exists in v3. The high tie rates observed later show that future work should freeze an ambiguity policy (for example, abstaining on unresolved top ties or requesting a user clarification) before examining new outcomes.

Robustness under tested conditions is represented by four observable mechanisms: database mutation denial, bounded execution, complete named-state eligibility, and deterministic failure handling. Candidate-order stability is assessed separately in Results. Algorithm 1 states the resulting reference-isolated coordination rule.

Algorithm 1 Reference-isolated deterministic coordination

Require: question x_i, schema S_i, external candidates $\mathcal{Y}_i$, read-only executor, named state evidence

```
post structured intent and schema grounding to the append-only blackboard
canonicalize and deduplicate externally produced candidates
for each candidate y ∈ Y_i do
    apply single-statement read-only validation
    record execution and available reference-free state evidence
end for
if required named-state coverage is incomplete for a candidate then
    mark that candidate ineligible (fail closed)
end if
if no safe executable candidate meets all registered gates then
    abstain
else
    apply the fixed validation, state-evidence, coverage, and order rule
end if
post the decision and seal the blackboard
load reference results only for offline evaluation
```

3.6. Generation Study

The paired design crosses context package $c \in \{0, 1\}$ and composite hint $h \in \{0, 1\}$. F00 uses full DDL and global values without the hint; F01 adds the hint; F10 uses compact domain-grounded context without the hint; and F11 combines compact context and the

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hint. The same 180 questions appear in all four cells for each backbone, yielding 720 Qwen and 720 Granite predictions.

Because each question appears in every prompt cell, cell differences are paired within question. The two backbone snapshots form parallel finite-corpus experiments, not independent replications drawn from a model population. Difficulty labels and normalized-SQL clusters describe the corpus structure. Cluster-based randomization reduces the influence of near-duplicate SQL forms, while composition-sensitivity intervals show how estimates vary when observed clusters are resampled. Neither procedure converts this development-visible synthetic set into a population sample.

Qwen2.5-Coder-7B Q4_K_M and Granite-3.3-8B Q4_K_M were run at temperature zero with fixed snapshots. Each prompt produced exactly one response. The parser retained the first SQL candidate through the first semicolon; the validator then applied the read-only policy. There was no candidate ranking, multi-agent negotiation, or repair in this experiment.

Strict execution equality is primary. A common-target projected-column indicator is a secondary manipulation check. Paired finite-set contrasts use normalized-SQL clusters as dependence proxies, registered sign-flip tests, composition-sensitivity bootstrap intervals, and Holm correction [35–37]. The intervals describe sensitivity to the observed corpus composition, not population confidence.

3.7. Component, Constructed-State, and Public-Database Studies

The separately specified component study contains 700 scored calls. E1 compares candidate generation with and without presented value evidence on 170 paired questions in 61 groups. E2 requests up to three candidates and compares a deterministic reference-free selector with the first candidate on 180 questions in 70 groups. Question-level paired differences are averaged with equal question weights; bootstrap resampling and sign assignments operate at group level. Selection is sealed before gold loading. Rescue, harm, and oracle-at-three are descriptive; oracle-at-three is a gold-only upper-bound diagnostic. The three two-member Holm families are E1 across backbones, E2 across backbones, and the two cross-backbone modifiers.

The constructed-state reliability protocol reproduced all 1440 T0 labels and then executed every prediction over 18 states. The primary 66-question analysis uses a question-weighted logical-AND endpoint over 15 semantic states. A prediction passes only when it agrees with the reference on every included state. The nine specified contrasts form a separate Holm family; with only 12 structural groups, all $2^{12} = 4096$ group-sign assignments are enumerated. The sharp-null interpretation assumes exchangeability of those 12 group signs. The small group count limits randomization resolution, and the groups are dependence proxies rather than sampled deployment clusters.

The BIRD protocol compares direct prompting, decomposition, schema selection, and bounded execution repair. Each method has a fixed call pattern and zero retries. Accuracy and method differences retain equal item weighting over 500 items; pairwise randomization assigns one common sign to all paired item differences within each of 11 databases and applies Holm correction across 12 comparisons. Its sharp-null interpretation assumes database-level sign exchangeability. With only eleven clusters, the attainable evidence is coarse, and the databases do not represent a random sample of power-grid systems.

The four protocols answer different questions and are not pooled. GridDB factorial cells test prompt-package manipulations on a controlled grid schema. E1 and E2 test presented values and reference-free selection. The constructed-state study tests agreement across transformed databases, and BIRD tests non-grid workflow portability.

3.8. Historical-Pool Selection Study

The replay verifies SHA-256 for two 720-row prediction ledgers and the 25,920-row constructed-state ledger before reading them. For each question it pools the eight existing outputs from two backbones and four prompt cells, canonicalizes SQL, and removes duplicates. T0 fields provide execution and metadata-shape evidence. The Critic records named-state evidence as unavailable because existing state-agreement labels are gold-relative.

This preliminary replay is a coverage diagnostic. The candidates were generated under different models and prompt packages, and no selected output from this step is used as a performance estimate.

The historical-pool selection study reuses exactly eight source slots for each of the 180 GridDB evaluation questions: four Qwen prompt cells followed by four Granite prompt cells. It performs no LLM call and creates no new question or SQL label. Before selection, all three methods share one precomputed eight-by-four execution ledger: T0 plus three metamorphic witnesses for every slot. The shared ledger contains 5760 physical evidence-collection attempts; this is not a separate runtime cost assigned to each selector. The fixed-order control ignores the ranking evidence and chooses slot 0. Validation-only selection ranks candidates without named-state evidence, whereas complete-witness selection requires all three reference-free metamorphic outcomes. Incomplete coverage or no passing candidate causes abstention.

The three query-blind witnesses have distinct SHA-256 values and are built from T0 without reading questions, prompts, predictions, scores, or gold. M1 adds an irrelevant relation and rows; M2 adds harmless indexes and rebuilds storage; M3 adds nullable probe columns to three business tables. M1 and M2 preserve original-schema denotation. M3 preserves explicitly projected original columns but deliberately allows wildcard projection to fail. Query order is preserved when candidate SQL contains `ORDER BY`; otherwise rows are compared as a multiset. The selector sees only question text extracted from retained prompt ledgers, the schema, eight SQL strings, executor traces, and T0-to-witness equivalence. It does not see gold SQL, answer-shape metadata, correctness, difficulty, feature tags, required literals, or gold-relative constructed-state labels.

These witnesses test metamorphic relations rather than alternative ground-truth answers. M1 asks whether an irrelevant table changes behavior, M2 whether logically neutral storage changes alter the result, and M3 whether projection is stable when nullable columns are appended. A pass means that the observed result satisfies the registered relation; it does not mean the original query expresses the user's intent. A fail can reveal brittleness, as with wildcard projection, but may also reflect a transformation outside the intended query equivalence class. For this reason, witness outcomes enter eligibility only under the disclosed construction and remain reference-free diagnostics.

Candidate order is a first-class experimental variable because the controller uses stable order for equal scores. Original order places four Qwen slots before four Granite slots and follows the prompt-cell sequence within each backbone. A post-review order audit changes only this tie breaker after the same evidence has been collected and exactly enumerates all $8! = 40{,}320$ global slot orders. It is a direct diagnostic of unresolved ranking information, not an alternative model or a new sample. The observed range motivates an abstention or clarification policy for future work.

Each selector used the same 1440 source slots and 5760 attempts, and no failed candidate was replaced or retried. The complete software-test matrix, protocol chronology, manifest hashes, and excluded predecessor records are provided in Supplementary Sections S1–S3.

Descriptive endpoints are strict execution accuracy over all 180 questions, coverage and abstention, complete-three-witness invariance of the selected candidate, and rescue/harm relative to C000. The unified-evaluator order audit reports the exact correct-count distribution across all global slot orders. A descriptive risk–coverage curve covers questions whose top-score tie set is no larger than k, for $k = 1, \ldots, 8$, and uses the frozen original order within covered ties. Because same-item outcomes were accessible before these diagnostics were finalized, neither the permutation audit nor the risk–coverage curve is a confirmatory analysis.

A post-review reconciliation freezes evaluator MA-SQLGrid-GridDB-T0-shape-denotation-v1 and applies it to all eight fixed slots and the sealed selector choices. Candidate and gold SQL execute under the same bounded read-only T0 snapshot. Correctness requires successful execution, the frozen expected column count, and duplicate-preserving row equality (ordered when specified; floating absolute tolerance 10^{-6}). Empty–empty rows are compared only after both shape gates pass. SQL canonicalization is used for artifact identity, not correctness. This reconciliation is post-result evidence rather than preregistration.

3.9. Statistical Analysis

The statistical unit is the question, and every question receives equal weight. The principal estimand is the mean paired difference on the observed finite corpus. Normalized-SQL groups for GridDB and databases for BIRD are dependence proxies used for grouped resampling and sign assignment; they are not random samples from a deployment population. Composition-sensitivity intervals describe how a finite-corpus estimate changes when the observed groups are reweighted and are not population confidence intervals.

The randomization procedures assume exchangeability of group-level signs under the tested sharp null. Exact enumeration is used for the 12-group constructed-state analysis ($2^{12} = 4096$ assignments), while the larger registered families use the documented Monte Carlo procedures and seeds. All tests are two-sided, zero paired differences remain in the estimand and contribute no sign, and the Holm families are defined separately for the nine GridDB factorial effects, projected-column manipulation checks, three two-member component families, nine constructed-state contrasts, and 12 BIRD pairings. Count differences, paired effects, rescue/harm, and sensitivity results remain visible whether or not a Holm-adjusted decision is met.

The historical-pool comparison is reported through complete counts, paired rescue/harm, coverage, abstention, tie multiplicity, and candidate-order sensitivity. Because same-item outcomes were accessible during earlier development, these quantities describe the observed pool and do not estimate a population or causal multi-agent effect.

3.10. Implementation and Reproducibility

The local generation experiments used the identified Qwen2.5-Coder-7B-Instruct Q4_K_M and Granite-3.3-8B-Instruct Q4_K_M snapshots under frozen protocols. The BIRD formal protocol fixed Python 3.10.11, SQLite 3.40.1, model order, prompts, evaluator, and zero retries. The five-role controller, read-only executor, state-evidence gate, and deterministic replay have regression and adversarial tests. Main-text results always retain the control version actually used to generate them; later executor corrections are reported as implementation evidence and are not applied retroactively.

Detailed computational records, artifact hashes, test inventories, execution history, and rights information are provided in the Supplementary Materials.

https://doi.org/10.3390/app1010000

3.10.1. Generative-AI Assistance and Author Responsibility

The local SQL-generation experiments used the identified Qwen2.5-Coder-7B-Instruct Q4_K_M and Granite-3.3-8B-Instruct Q4_K_M snapshots under the protocols described above. During manuscript preparation, the authors also used the OpenAI Codex agent interface for prose editing, experimental suggestions, Python/LaTeX code review, computational verification, and local build and visual-quality checks. The exact backend model identifier and version were not retained for every session.

All executable suggestions were run locally against retained inputs, and every reported value was obtained from retained or independently recomputed records rather than generated prose. The authors verified the analyses, interpretations, citations, declarations, and final text and retain full responsibility for the manuscript. Codex output was not treated as qualified power-system expert annotation or independent adjudication. External machine-generated question–SQL candidates remain labeled machine silver. All six scientific figures are deterministic code-native graphics; no generative-image output is used as quantitative evidence.

4. Results

4.1. Overview of Empirical Findings

The experiments produced three main observations. First, schema context, structural hints, and presented values affected the two model snapshots differently; no single context strategy dominated across every analysis. Second, under the unified evaluator, historical-pool reference matches were 76/180 for C000, 99/180 for validation ranking, and 100/180 with complete witness evidence, while the fixed Qwen F01 source reached 129/180. Third, both evidence-aware selectors produced top-score ties on 130/180 questions and remained sensitive to candidate order. Validation evidence changed choices relative to C000, but the available signals often remained too coarse for stable ranking and did not outperform the best fixed source.

The results are organized by research question. GridDB and BIRD address context and portability; the historical-pool analysis addresses evidence-aware selection; and the constructed-state, tie, and order analyses address stability. Implementation conformance is summarized after the empirical results and documented fully in the Supplementary Materials.

4.2. Context and Component Effects on SQL Generation

All 1440 scheduled predictions are present, and independent SQLite re-execution reproduced the stored execution verdicts. Table 4 reports strict execution equality, and Figure 2 visualizes the same audited cell estimates.

Table 4. Strict execution equality over 180 GridDB questions per cell.

Backbone	F00 full/no hint	F01 full/hint	F10 compact/no hint	F11 compact/hint
Qwen	76/180 (0.4222)	129/180 (0.7167)	78/180 (0.4333)	108/180 (0.6000)
Granite	77/180 (0.4278)	100/180 (0.5556)	74/180 (0.4111)	108/180 (0.6000)

The common-target projected-column diagnostic is 0.5222 in F00 for both backbones. Qwen obtains 0.9667, 0.5889, and 0.9611 in F01, F10, and F11; Granite obtains 0.8778, 0.5667, and 0.9222. These values show that returning the requested number of columns is easier than returning the intended rows. In Qwen F01, projected-column conformity is 0.9667 while execution equality is 0.7167.

The eight cells also show that no single prompt summarizes the design. Adding the hint to full context increases the point estimate for both backbones, whereas adding compact

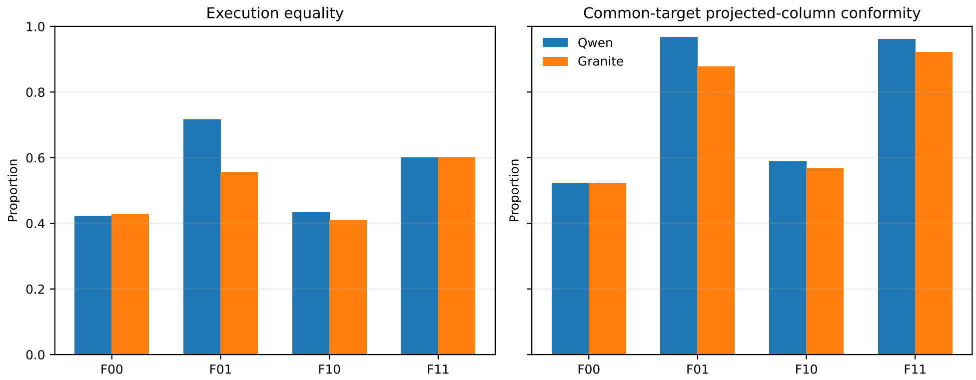

Figure 2. Audited GridDB cell point estimates. Each cell contains all 180 attempts; the figure is descriptive and does not replace registered factorial contrasts.

context without the hint leaves execution equality close to F00. Under compact context, the hint produces 108 correct answers for both models, but this numerical convergence arises from different unhinted baselines and does not establish identical behavior. Reporting every cell preserves these interactions instead of selecting only the highest value.

The 0.2500 gap between Qwen F01 projected-column conformity and strict execution equality warns against structural proxy endpoints. A model may recognize that one column, an aggregate, or an ordered row is requested while still choosing the wrong table, join, predicate, value, or boundary. Shape diagnostics can localize errors and inform adjudication, but they cannot replace result equality or qualified semantic review.

4.2.1. Factorial Effects and Multiplicity

Table 5 reports the complete primary family. For Qwen, the composite-hint execution effect is +0.2306, the package–hint interaction is −0.1278, and the compact-package main effect is −0.0528. Granite's corresponding estimates are +0.1583, +0.0611, and +0.0139. None of the nine primary execution tests survives Holm correction. Qwen's hint effect has raw/adjusted $p = 0.01372/0.09604$, its interaction 0.00919/0.07352, and the cross-backbone hint modifier 0.00640/0.05760. Accordingly, no primary factorial execution effect or modifier is described as statistically nonzero.

Table 5. Complete primary factorial family for strict execution equality. Estimates are question-weighted finite-set differences; intervals are 95% normalized-SQL-group composition-sensitivity intervals. Each sign-randomization used 70 groups, 100,000 draws, and the displayed raw values were Holm-adjusted jointly over all nine rows.

Scope	Effect	Estimate	Composition interval	Raw p	Holm p
Qwen	Compact-package main	-0.0528	[-0.1131, 0.0000]	0.10523	0.52614
Qwen	Structural-hint main	0.2306	[0.0548, 0.4355]	0.01372	0.09604
Qwen	Package × hint	-0.1278	[-0.2454, -0.0341]	0.00919	0.07352
Granite	Compact-package main	0.0139	[-0.0584, 0.0798]	0.85021	1.00000
Granite	Structural-hint main	0.1583	[-0.0193, 0.3662]	0.15480	0.61919
Granite	Package × hint	0.0611	[-0.0539, 0.2013]	0.55890	1.00000
Granite–Qwen	Backbone × package	0.0667	[-0.0360, 0.1814]	0.41478	1.00000
Granite–Qwen	Backbone × hint	-0.0722	[-0.1229, -0.0276]	0.00640	0.05760
Granite–Qwen	Three-way interaction	0.1889	[0.0090, 0.4321]	0.07010	0.42060

The normalized-SQL mapping contains 70 groups, 58 singletons, and a maximum group size of 19. Question-level weighting is retained; groups are resampling and sign-assignment proxies, not verified authoring templates or sampled clusters. The randomization interpretation requires exchangeability of group signs under the tested sharp null, while the composition interval asks how the finite-corpus mean changes when those observed groups are reweighted. Neither assumption follows from the data, and the high singleton fraction limits protection against unmeasured dependence.

The displayed composition intervals are pointwise sensitivity intervals; they are neither simultaneous/familywise intervals nor inversions of the nine Holm-adjusted randomization tests. An interval that excludes zero can therefore coexist with a Holm non-rejection. The multiplicity decision, not interval exclusion, controls statements about the registered nine-test family.

No minimum practically important execution-accuracy difference, equivalence margin, or non-inferiority margin was registered. The study therefore cannot infer practical equivalence from a non-rejection, and the 1/180 complete-witness difference is a mechanism trace rather than evidence of a meaningful benefit.

In the separately corrected secondary family, the common-target hint effects survive: +0.4083 for Qwen (adjusted $p = 0.000090$) and +0.3556 for Granite (adjusted $p = 0.01944$). Because the hint supplies structural information later scored by this endpoint, these are manipulation-check results, not independent semantic evidence.

The multiplicity result changes the emphasis of the factorial study. Large raw contrasts motivate hypotheses about structural instructions, but none of the nine primary execution comparisons meets the registered familywise rule. The projected-column findings verify that the manipulation changed a closely aligned surface property. They do not show that the compact package or hint produces a backbone-independent accuracy gain. Future work should separate schema pruning, value presentation, normalization, and structural hints rather than retaining composite packages.

4.2.2. Presented Values and Candidate Selection Components

The component run completed all 700 scored calls under a prospectively frozen call-and-selection procedure on development-visible GridDB items. On the paired 170-question subset, Qwen first-candidate correctness increased from 83/170 to 101/170, a +0.1059 difference (95% composition-sensitivity interval [+0.0282, +0.2013], adjusted $p = 0.0310$). Granite remained 69/170 in both paired conditions, with an interval spanning zero. The cross-backbone modifier did not meet the registered rule, so the Qwen result is not replicated across backbones and is not a five-role result.

The deterministic selector changed 24 Qwen and 36 Granite choices. Relative to the first candidate, it rescued eight and harmed one Qwen question, and rescued ten while harming none for Granite. Execution effects were +0.0389 and +0.0556; neither survived its two-test Holm family. Oracle-at-three values of 0.6389 and 0.4667 are gold-only diagnostics and are not deployable selector results.

For Qwen, the value-evidence intervention adds 18 correct first candidates when the 170 eligible V0 questions are compared with their V1 counterparts under the registered analysis; the corresponding Granite estimate is zero. The disagreement is more informative than averaging the backbones: the effect depends on how a frozen model uses schema and value context. Oracle-at-three measures candidate-pool headroom, whereas the reference-free selector measures how much of that headroom the recorded rule can recover without gold.

Complete component counts appear in the numerical supplement. On all 180 V1 questions, Qwen first/selector/oracle-at-three counts were 105/112/115, and Granite counts were 71/81/84; the paired 170-question intervention counts and all registered effects appear in Table 6. These are component, not five-role, outcomes.

The three two-member Holm families are E1 across backbones, E2 across backbones, and Cross-E across modifiers. The Granite E2 pointwise interval excludes zero while its adjusted decision does not meet the registered rule; these summaries are not contradictory.

Table 6. Complete component-effect families. Intervals are pointwise group-composition sensitivity summaries, not simultaneous intervals and not inversions of the Holm tests. Each family contains the two displayed members indicated by its identifier.

Family	Contrast	N	Groups	Estimate	Composition interval	Raw p	Holm p
E1	Qwen V1–V0 first candidate	170	61	0.1059	[0.0282, 0.2013]	0.01552	0.03104
E1	Granite V1–V0 first candidate	170	61	0.0000	[-0.1902, 0.1705]	1.00000	1.00000
E2	Qwen selector–first V1	180	70	0.0389	[-0.0081, 0.1071]	0.50080	0.50080
E2	Granite selector–first V1	180	70	0.0556	[0.0075, 0.1232]	0.06425	0.12850
Cross-E	Granite–Qwen E1 effects	170	61	-0.1059	[-0.2701, 0.0394]	0.26986	0.53971
Cross-E	Granite–Qwen E2 effects	180	70	0.0167	[-0.0062, 0.0457]	0.37477	0.53971

The 61- and 70-group mappings retain question weights but are dependence proxies, not a probability sample. Figure 3 visualizes these effects without replacing the adjusted decisions.

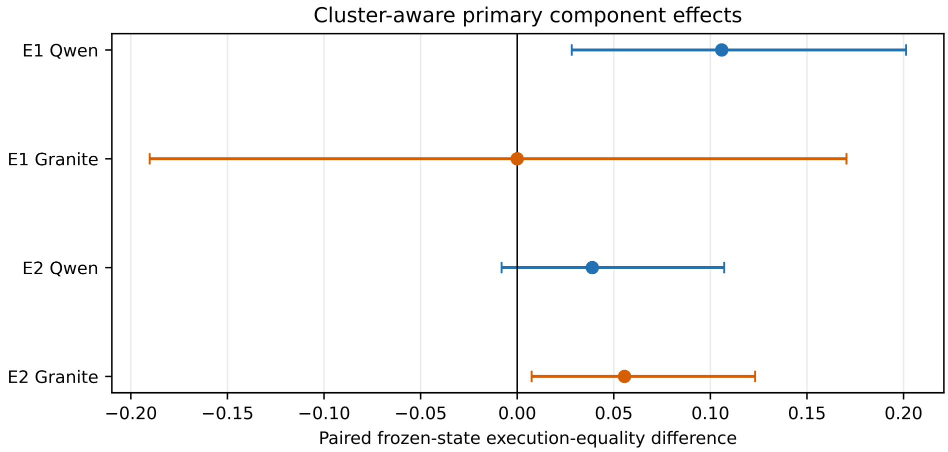

Figure 3. Component effects under a prospectively frozen call-and-selection procedure on development-visible GridDB items. Intervals describe composition sensitivity; the registered decision also requires multiplicity-adjusted evidence.

4.3. Constructed-State Stability

The v5 scorer produced all 25,920 protocol-specified rows, and its T0 slice reproduced 1440/1440 snapshot labels. Table 7 and Figure 4 report the 66-question automatic subset. The 15-state logical-AND rates range from 0.6212 (Granite F01) to 0.8182 (Granite F00). All nine Holm-adjusted values equal 1.0000 under exact cluster-sign enumeration. No context, hint, interaction, or cross-backbone multi-state effect meets the declared rule. These rates characterize constructed-state agreement, not population accuracy or human-certified equivalence.

The ordering under this endpoint differs from the single-snapshot table. Granite F00 has the highest logical-AND rate on the 66-question automatic subset even though Granite F11 has the larger T0 execution count over all 180 questions. The denominators and endpoints differ, so this is not a contradiction. The reversal shows why a robustness statement must name both the admitted question subset and the required state set.

Table 7. Complete 15-state logical-AND counts for the prediction-blinded 66-question automatic subset.

Backbone	F00	F01	F10	F11
Qwen	49/66 (0.7424)	48/66 (0.7273)	53/66 (0.8030)	43/66 (0.6515)
Granite	54/66 (0.8182)	41/66 (0.6212)	51/66 (0.7727)	49/66 (0.7424)

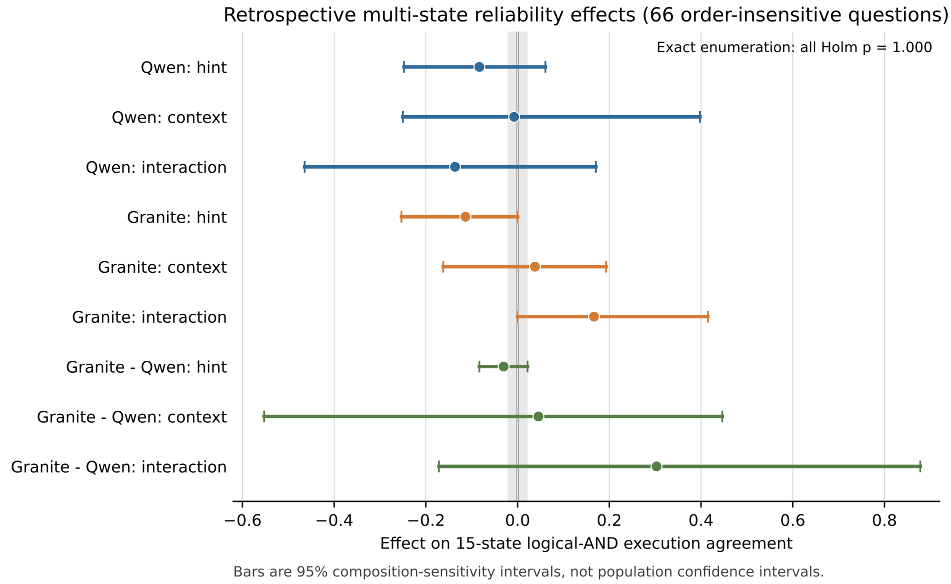

Figure 4. GridDB multi-state logical-AND agreement for the prediction-blinded automatic subset. The endpoint is a constructed-state robustness diagnostic.

4.4. Cross-Database Portability on BIRD

Table 8 reports all eight cells. BIRD Mini-Dev is a public non-grid portability benchmark and does not validate power-grid semantics. B0–B2 use one physical generation call per item; B3 uses two, so the inherited comparison estimates frozen workflow efficacy under unequal call counts rather than a call-matched repair effect. Each backbone completed 2500 calls, 2000 final predictions, zero retries, and zero final-ledger omissions. Latency and token totals are unavailable in the retained summary and were not reconstructed.

Table 8. Inherited BIRD Mini-Dev v1.1 results (non-grid portability evidence; 11 database clusters). Intervals are database-cluster composition-sensitivity intervals.

Backbone	Method	Correct/500	Accuracy	Interval	Calls/item	Final-ledger omissions
Qwen	B0 direct	189/500	0.378	[0.286, 0.470]	1	0
Qwen	B1 decomposition	151/500	0.302	[0.203, 0.401]	1	0
Qwen	B2 schema selection	197/500	0.394	[0.279, 0.511]	1	0
Qwen	B3 execution repair	174/500	0.348	[0.250, 0.452]	2	0
Granite	B0 direct	102/500	0.204	[0.125, 0.284]	1	0
Granite	B1 decomposition	105/500	0.210	[0.131, 0.294]	1	0
Granite	B2 schema selection	101/500	0.202	[0.137, 0.273]	1	0
Granite	B3 execution repair	118/500	0.236	[0.159, 0.318]	2	0

After Holm adjustment, Qwen decomposition is 0.076 below direct prompting (adjusted $p = 0.0430$), and schema selection is 0.092 above decomposition (adjusted $p = 0.0117$). No Granite contrast survives correction. Different method orders across the two model snapshots argue against a universal prompting recipe. Two incomplete predecessor runs were retained for traceability and excluded from all reported denominators; their disposition is documented in the Supplementary Materials.

The BIRD results also constrain the meaning of "agentic" improvement. Decomposition reduces Qwen accuracy relative to direct prompting, while bounded repair uses twice the physical calls without becoming the best Qwen method. Granite shows a different descriptive ordering and no adjusted contrast. More stages or calls do not guarantee a better outcome. A future comparison must count physical generations, retries, context tokens, and failed attempts, not only final predictions.

4.5. Validation-Aware Selection in the Historical Candidate Pool

The replay covered all 180 GridDB questions without new generation. At least two distinct SQL candidates were available for 173 questions, and 172 had at least two eligible candidates after safety and execution checks. Seven questions had only one unique candidate, and one had fewer than two eligible candidates. These counts describe interface coverage; the full candidate-multiplicity distribution is reported in the numerical supplement.

4.5.1. Reference Matches, Coverage, and Rescue/Harm

The deterministic descriptive re-execution retained 5760 candidate–state attempts; 332 were non-executable or authorization/resource/SQL failures and remained in the all-attempt ledger. Every question nevertheless had at least one eligible candidate, so both adjudicating selectors covered 180/180 questions and abstained on 0/180. This zero-abstention result is a consequence of at least one eligible slot plus forced stable-order resolution of post-ranking ties, not confidence-calibrated coverage. Table 9 reports the recorded rules.

Keeping all 332 failures matters for two reasons. Removing them would overstate the executability of the historical pool, and retaining their slot identities verifies that selectors did not receive replacements or retries. Full question-level coverage means only that another slot was eligible for every question; it does not mean every candidate was valid or that an operational system should always answer.

Table 9. Unified-evaluator comparison for the historical eight-slot pool. C000 is byte-identical in normalized SQL to Qwen F00 for all 180 questions. Differences, paired question-composition intervals, rescue/harm counts, and exact discordant-sign tests use C000 as baseline; Holm adjustment covers nine unique comparisons. This reconciliation is post-result.

Method	Correct/180	Accuracy	Difference	95% comp. interval	Rescue/harm	Holm p
C000 / Qwen F00	76/180	0.4222	0	[0, 0]	0/0	–
Qwen F01	129/180	0.7167	+0.2944	[+0.2222, +0.3667]	56/3	< 0.000001
Qwen F10	78/180	0.4333	+0.0111	[-0.0111, +0.0333]	3/1	1.000000
Qwen F11	108/180	0.6000	+0.1778	[+0.1000, +0.2556]	46/14	0.000211
Granite F00	77/180	0.4278	+0.0056	[-0.0278, +0.0444]	6/5	1.000000
Granite F01	100/180	0.5556	+0.1333	[+0.0500, +0.2222]	45/21	0.017090
Granite F10	74/180	0.4111	-0.0111	[-0.0500, +0.0278]	6/8	1.000000
Granite F11	108/180	0.6000	+0.1778	[+0.1056, +0.2500]	42/10	0.000054
Validation-only selector	99/180	0.5500	+0.1278	[+0.0778, +0.1833]	25/2	0.000040
Complete-witness selector	100/180	0.5556	+0.1333	[+0.0778, +0.1889]	26/2	0.000024

Qwen F00 and historical C000 contain the same 180 SQL artifacts. Their previously reported 76-versus-80 discrepancy came from evaluator policy: the historical row-only comparator treated Q104, Q107, Q110, and Q140 as matches because both results were empty, whereas all four candidate projections had the wrong column count. The unified evaluator applies the frozen shape gate before empty-row equality and reproduces all 1440 canonical fixed-slot verdicts. Under that evaluator, validation-only and complete-witness selection differ by one match and remain 30 and 29 matches below Qwen F01, respectively.

Figure 5 juxtaposes reference matches and witness invariance. The one-match selector increment focuses interpretation on validation ranking and tie policy, not a broad witness benefit.

Both evidence-aware selectors produced multiple highest-scoring candidates on 130/180 questions, with mean top-tie sizes of 5.4000 and 5.3889. With original-order tie breaking, 178/180 selections came from Qwen slots, reflecting source order rather than learned preference.

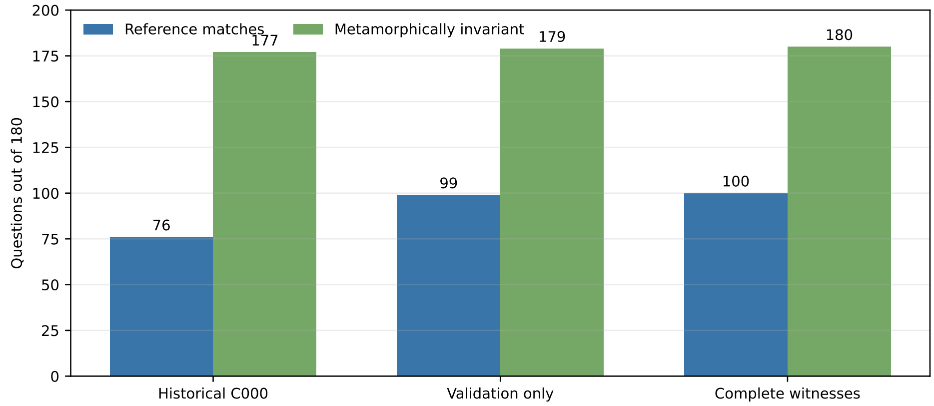

Figure 5. Selector endpoints over 180 historical-pool questions. Reference matches are loaded only after board sealing; invariance denotes agreement across the three constructed witnesses.

Most apparent source preference is mechanically induced. When all eight candidates tie, the first slot is selected regardless of backbone quality; when seven tie, the earliest surviving slot has the same advantage. The observed 178 Qwen selections measure how often the available reference-free features failed to separate candidates before the stable-order rule acted, not a learned preference for one backbone.

The numerical supplement reports the complete multiplicity and selected-slot distributions, including all eight-candidate ties.

Validation-only ranking added 23 net reference matches relative to C000 through 25 rescues and 2 harms. Complete-witness selection changed the chosen candidate on only Q039, yielding one further match (100/180 versus 99/180), one additional rescue, no additional harm, and no coverage or abstention change. Table 10 gives the constructed trace for this sole selector difference. Neither selector exceeded the fixed Qwen F01 source at 129/180.

4.6. Tie Structure, Order Sensitivity, and the Q039 Case

Table 10. Q039 is the sole validation-only versus complete-witness selection difference. "Match" denotes agreement with the synthetic reference.

Field	Validation-only	Complete witnesses	Interpretation
Request	"Show the first scheduled work order after June 2024."		Status/date ambiguity
Projection	`SELECT *`	`SELECT work_order_id, scheduled_date`	Wildcard vs. explicit fields
Shared filter	`scheduled_date > '2024-06-01' ORDER BY scheduled_date ASC LIMIT 1`		Ambiguous time scope
M3 outcome	Wildcard width changes	Two-column projection preserved	Projection stability
Reference result	No match	Match	One-question trace

M3 added nullable columns to `work_orders`; consequently, complete-witness eligibility rejected the wildcard projection and retained the explicit `work_order_id, scheduled_date` projection. The full SQL strings are supplied in the numerical supplement. Neither query resolves the request's status and date-boundary ambiguity through qualified review, so the extra reference match is interpreted as a projection-stability trace rather than an engineering correctness judgment.

Table 11 replaces the non-comparable historical row-only sensitivity values with an exact audit under the unified evaluator. The audit enumerated all $8! = 40{,}320$ global slot orders while preserving each question's frozen evidence scores. Validation-only selection ranged from 95 to 128 matches and complete-witness selection also ranged from 95 to 128; both had a median of 107. The original orders produced 99 and 100 matches, respectively, while the reverse order produced 116 for both selectors. Even the outcome-exposed maximum remained below the fixed Qwen F01 source at 129/180.

These permutations are diagnostics, not candidate policies: no order is preferred after outcomes are observed. A second descriptive audit abstained whenever the top-score set was not unique. It covered only 50/180 questions and matched 24 of them (0.4800), so top-tie size alone was not a calibrated confidence signal. The range across global orders and the poor strict-abstention result motivate a pre-registered semantic tie or abstention rule evaluated on an untouched pool.

A transparent post-run hash audit emitted no SQL text and found duplicate normalized SQL in 154/180 pools. Mean validation and complete-witness top sets fell from 5.400 and 5.389 slots to 3.150 and 3.139 unique strings; the mean pool contained 5.322 unique strings. Deduplication reduces apparent multiplicity but does not remove ambiguity. The outcome-aware right-step tie-size AURC values, 0.4826 and 0.4795, are diagnostics rather than calibration results.

Table 11. Exact global candidate-order sensitivity under the unified shape-and-denotation evaluator. All 40,320 orders are outcome-aware diagnostics; original and reverse are two members of that enumeration.

Selector	Original	Reverse	Minimum	Median	Mean	Maximum
Validation-only	99	116	95	107	109.845	128
Complete witness	100	116	95	107	109.929	128

The post-review utilization audit then traced which role outputs actually enter the historical-pool selector. Query Analyst fields feed aggregation, order, and lexical-match validation features. By contrast, the Schema Cartographer's grounding is posted to the blackboard but is not passed to candidate generation, validation, the critic, or adjudication in this frozen driver. The candidate provider is an adapter for stored SQL and makes no model calls. Table 12 therefore reports score-component diagnostics rather than claiming a prospective removal of five independently active agents. Safety and executability gates remain active in every variant.

Table 12. Post-review role-utilization and single-component diagnostics under the unified evaluator. Changes, gains, and losses are relative to the stated parent and use the original candidate order.

Selector variant	Parent	Correct/180	Choices changed	Gains	Losses
Validation original	—	99	0	0	0
Without all Query-Analyst-derived score terms	Validation original	76	64	2	25
Without shape term	Validation original	99	0	0	0
Without order term	Validation original	77	49	1	23
Without lexical value-hit term	Validation original	99	14	1	1
Complete witness original	—	100	0	0	0
Without constructed-state gate	Complete witness	99	1	0	1
Without recorded-only schema grounding	Complete witness	100	0	0	0

The order feature accounts for most of the observed post-review change in this decomposition: removing it changes 49 validation choices and lowers the match count by 22. Removing lexical value hits changes 14 choices but exchanges one gain for one loss, while removing the shape term changes none. Removing constructed-state eligibility changes only Q039. These same-item, outcome-aware diagnostics localize behavior in the frozen pool; they do not estimate the benefit of an independently implemented role on unseen questions.

An automated evaluator-state taxonomy further separates execution failure, output-shape mismatch, and wrong denotation after both shape gates pass. C000 has 1 execution error, 85 shape mismatches, and 18 wrong denotations among its 104 failures. Validation-only selection has 0, 42, and 39, respectively; complete-witness selection has 0, 41, and 39. The best fixed Qwen F01 source has 5 execution errors, 1 shape mismatch, and 45 wrong denotations. Thus the frozen selector removes many inadmissible or shape-inconsistent candidates but leaves a larger share of its remaining failures beyond the evaluator's shape gate. This automatic taxonomy does not distinguish wrong status meaning, units, time boundaries, topology intent, or professional usefulness; those categories remain subject to independent expert review.

The numerical supplement contains all eight GridDB cells, 12 component endpoints, eight 15-state cells, the exact unified-evaluator order histogram, the descriptive risk–coverage curve, the 360-row item-level tie ledger, selected-origin counts, and the Q039 trace. The main-text findings use different questions, candidates, call patterns, and endpoints, so they are not combined into a single framework accuracy. Across these studies, the common diagnostic finding is that context and validation evidence can change observable behavior, while stable semantic selection remains unresolved.

4.7. Implementation Reliability

The five-role controller and executor were tested for typed handoffs, mutation denial, bounded execution, trace completeness, deterministic replay, and fail-closed state coverage. Failed attempts remain in the evidence ledger and cannot be silently replaced. The executor used to create the historical candidate and evidence ledgers enforced SQLite read-only mode, `query_only`, authorization callbacks, disabled extensions, and time, opcode, and row bounds. A later revision added cell-byte, total-result-byte, width, and function controls; these tests strengthen the implementation boundary but do not reinterpret the frozen ledgers. The unified evaluator subsequently scored the stored fixed and selected SQL under one explicit shape-and-denotation rule. The historical selector ledger contains 3960 blackboard messages and 5760 database attempts (5428 executable and 332 SQLite failures) across 953 distinct SQL hashes; it contains no selector-stage model calls. Recorded local attempt time totals 839 ms, but timer granularity places both the median and 95th percentile at 0 ms, so these values are an audit of the ledger rather than a deployment-latency estimate. Supplementary Tables S1–S3 provide the complete test inventory and protocol attribution.

5. Discussion

5.1. Model-Dependent Use of Context and Values

The GridDB and BIRD studies show that additional structure is useful only when a model can exploit it. Structural hints sharply changed projected-column adherence, yet none of the nine primary GridDB execution effects survived Holm correction. Presented values improved Qwen first-candidate matches on the paired component subset but produced no change for Granite. BIRD produced another model-specific ordering: schema selection was best for Qwen, whereas bounded repair had the highest Granite point estimate. These patterns are consistent with schema-linking research in which context selection trades recall against distraction [16–18]. More context is not uniformly better. Its effect depends on the model snapshot, the serialization, and the type of evidence added.

This dependence motivates separate Query Analyst and Schema Cartographer interfaces, which allow later studies to vary intent extraction, schema retrieval, or value presentation without changing the executor or controller. It does not show that the separation improves accuracy. Projected-column conformity is also not a semantic endpoint:

a query can adopt the requested result shape while retaining the wrong table, predicate, value, or join.

5.2. Why Validation-Aware Selection Changed the Historical Pool

Validation-aware ranking added 23 net reference matches relative to C000 through 25 rescues and 2 harms. The change is consistent with a simple mechanism: execution failures, result-shape inconsistency, ordering mismatch, and weak value evidence remove or demote candidates that a source-order rule would otherwise select. Because all selectors used the same precomputed ledger, the comparison isolates selection behavior within this pool rather than differences in candidate generation. It did not exceed the fixed Qwen F01 source, which added 53 matches relative to C000 under the same evaluator.

The descriptive result is useful for system design. It shows that execution and trace evidence can change which existing candidate is chosen, and it identifies the questions on which the change helps or harms. A prospective comparison must regenerate candidates under equal call and token budgets, but the present ledger supplies a concrete hypothesis and an auditable selector interface for that test.

The error decomposition narrows that hypothesis: automatic validation is strongest at rejecting execution and output-shape failures, not at resolving semantically competing denotations. A future selector should therefore be evaluated on whether it reduces wrong-denotation and expert-semantic errors without simply exchanging them for shape conformity.

5.3. Limited Increment from Constructed-State Evidence

Complete witness evidence added only one match beyond validation-only ranking. The three witnesses cover an irrelevant relation, storage/index changes, and nullable schema extension. These transformations detect structural or execution brittleness but contain little information about coded statuses, intended time boundaries, units, or business definitions. Their limited incremental effect is consistent with this design.

Q039 illustrates the distinction. The nullable-extension witness rejects a wildcard projection and preserves an explicit two-column projection, but neither candidate resolves whether "scheduled" denotes a status or a date and neither settles the meaning of "after June". Constructed states can expose a specific implementation weakness; richer semantic evidence or user clarification is needed to resolve the underlying intent.

5.4. Ties as the Main Ranking Bottleneck

Both adjudicating selectors tied at the highest score on 130/180 questions, with about 5.4 candidates in the average top-tie set. Under the unified evaluator, exact enumeration of all 40,320 global orders moved both selectors across a 95–128-match range; the complete-witness result changed from 100 to 116 under simple reversal. Strict no-tie abstention covered 50 questions but matched only 24. This sensitivity reveals a bottleneck that aggregate accuracy alone would hide: the evidence can eliminate some inferior candidates but often cannot rank the survivors, and tie multiplicity alone is not a calibrated uncertainty measure.

The next design problem is to obtain reference-free semantic evidence that breaks ties consistently, or to abstain when such evidence is unavailable; generating more candidates alone does not resolve the tie structure. Candidate-order rules, ambiguity margins, clarification requests, and risk–coverage targets should be fixed on development data before an untouched evaluation set is opened. The shared blackboard trace provides the observations needed to study each alternative without changing the upstream generation records.

5.5. Engineering Significance

MA-SQLGrid is a traceable research prototype for natural-language access to structured grid data. Developers can locate a failure at the intent, schema, candidate, execution, state, or selection stage instead of receiving only a final SQL string. Database-enforced read-only execution, resource limits, and complete-evidence gates also reduce several implementation risks that lexical SQL filtering alone does not address.

These mechanisms are a foundation for safer data access, not an operational authorization layer. An executable query can still return the wrong assets, omit time bounds, or misinterpret codes. Consequential use requires authenticated users, least-privilege views, institutional access policy, and inspection by qualified personnel. The scientific value of the architecture lies in making those remaining assumptions and errors visible and reproducible.

Figure 6 summarizes how the four empirical streams connect to the MA-SQLGrid workflow.

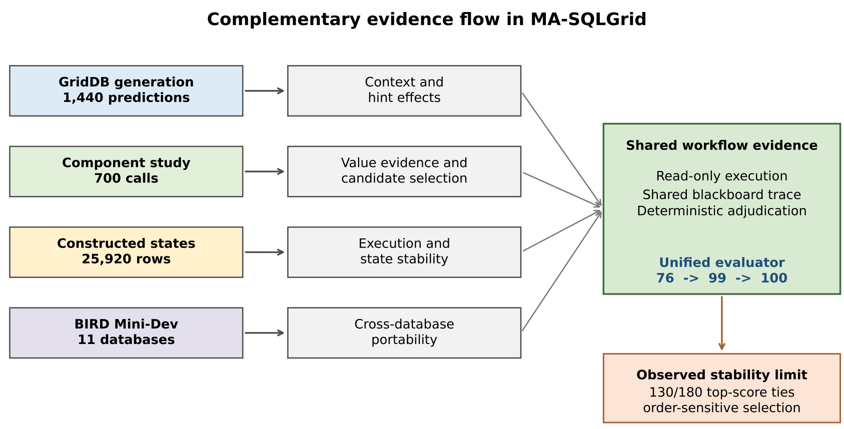

Figure 6. Evidence flow from generation studies to read-only validation, constructed-state analysis, and historical-pool selection. The four empirical streams answer complementary questions and are not pooled into one framework accuracy.

6. Limitations and Future Work

Data and external validity. GridDB is a synthetic database with eight tables, 98 rows, and 180 evaluation questions, and its evaluation partition was visible during development. BIRD broadens database coverage but is not a grid benchmark. RTS-GMLC, SimBench, and NERC-derived question–SQL assets remain machine-adjudicated silver resources. The next evaluation should use unseen grid schemas and queries independently reviewed by power-system and database specialists for projections, units, time semantics, ordering, and tie handling.

Experimental design. The historical candidate pool combines two backbones and four prompt cells whose outcomes had been examined during earlier development. It supports a descriptive selector comparison, not an end-to-end five-role effect. A new study should hold model snapshot, decoding, candidate count, physical calls, token budget, executor, and evaluator constant across a direct single-candidate baseline, staged one-candidate handoffs, validation-aware multi-candidate selection, and the same selection design with registered state evidence.

Semantic validity. Read-only execution and constructed states test admissibility, executability, and selected invariances. They do not establish that a query reflects a grid

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professional's intended business meaning. Future work should combine expert review with reference-free evidence for units, status definitions, time windows, topology relations, and result granularity. Model- or author-assisted triage should remain distinct from independent expert adjudication.

Ranking stability. The high tie rate and candidate-order sensitivity show that the current evidence score is weak when several candidates execute successfully. Future protocols should calibrate semantic ranking, abstention margins, and clarification policies on development-only data and report risk–coverage on an untouched evaluation set. A trained schema linker and low-confidence full-schema fallback can be introduced within the same trace interface, with every change receiving a new protocol identity.

7. Conclusions

MA-SQLGrid frames power-grid Text-to-SQL as a traceable five-role multi-agent decision process. The role interfaces separate intent analysis, schema mapping, candidate normalization, read-only execution validation, and constructed-state evidence. A deterministic controller records their handoffs through a shared blackboard trace, selects only eligible candidates, and can abstain when required evidence is incomplete. This architecture provides an auditable interface for locating errors that would be hidden by a final-SQL-only pipeline.

The experiments show that context and value evidence depend on the model and evaluation setting. Under the unified evaluator, the historical C000 policy produced 76/180 reference matches, validation-aware ranking produced 99/180, and complete witness evidence produced 100/180. These descriptive gains of 23 and 24 characterize selection within the pool, not an architectural treatment effect; neither selector exceeded the fixed Qwen F01 source at 129/180. Top-score ties on 130/180 questions and the exact 95–128 range across global candidate orders expose an unresolved semantic-ranking problem.

MA-SQLGrid provides an auditable research interface for power-grid Text-to-SQL. Claims about broader efficacy or deployment require a call- and budget-matched evaluation on unseen, independently expert-reviewed grid queries, with pre-registered semantic evidence, tie handling, abstention, and risk–coverage analysis.

Supplementary Materials: The Supplementary Materials include software and executor tests, protocol and version records, complete numerical tables, figure provenance, rights information, and reproducibility instructions.

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Data Availability Statement: The manuscript-bound, rights-safe technical package can be inspected in the public powergrid_benchmark repository under the frozen tag cmc-2026-08-24-v2 via the frozen MA-SQLGrid package. It includes the numerical supplement, code, tests, derived evidence, and reproducibility records used for this version; BIRD Mini-Dev remains available from its public provider. Owing to third-party licensing restrictions, raw GridDB and BIRD databases and restricted source records are not redistributed, and source-dependent RTS-GMLC or SimBench artifacts may be provided for editorial or peer-review verification only when permitted by the applicable licenses. The historical `ma-sqlgrid` repository is not the manuscript-bound release for this version.

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References

1. Yu, T.; Zhang, R.; Yang, K.; Yasunaga, M.; Wang, D.; Li, Z.; Ma, J.; Li, I.; Yao, Q.; Roman, S.; et al. Spider: A Large-Scale Human-Labeled Dataset for Complex and Cross-Domain Semantic Parsing and Text-to-SQL Task. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, 2018, pp. 3911–3921. https://doi.org/10.18653/v1/D18-1425.
2. Li, J.; Hui, B.; Qu, G.; Yang, J.; Li, B.; Li, B.; Wang, B.; Qin, B.; Geng, R.; Huo, N.; et al. Can LLM Already Serve as a Database Interface? A Big Bench for Large-Scale Database Grounded Text-to-SQLs. In Proceedings of the Advances in Neural Information Processing Systems 36, 2023.
3. Lei, F.; Chen, J.; Ye, Y.; Cao, R.; Shin, D.; Su, H.; Suo, Z.; Gao, H.; Hu, W.; Yin, P.; et al. Spider 2.0: Evaluating Language Models on Real-World Enterprise Text-to-SQL Workflows. In Proceedings of the The Thirteenth International Conference on Learning Representations, 2025.
4. Li, C.; Wang, Y.; Wu, Z.; Yu, Z.; Zhao, F.; Huang, S.; Dai, X. MultiSQL: A Schema-Integrated Context-Dependent Text-to-SQL Dataset with Diverse SQL Operations. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2024, 2024, pp. 13857–13867. https://doi.org/10.18653/v1/2024.findings-acl.823.
5. Wang, B.; Shin, R.; Liu, X.; Polozov, O.; Richardson, M. RAT-SQL: Relation-Aware Schema Encoding and Linking for Text-to-SQL Parsers. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, 2020, pp. 7567–7578. https://doi.org/10.18653/v1/2020.acl-main.677.
6. Scholak, T.; Schucher, N.; Bahdanau, D. PICARD: Parsing Incrementally for Constrained Auto-Regressive Decoding from Language Models. In Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, 2021, pp. 9895–9901. https://doi.org/10.18653/v1/2021.emnlp-main.779.
7. Nan, L.; Zhao, Y.; Zou, W.; Ri, N.; Tae, J.; Zhang, E.; Cohan, A.; Radev, D. Enhancing Text-to-SQL Capabilities of Large Language Models: A Study on Prompt Design Strategies. In Proceedings of the Findings of the Association for Computational Linguistics: EMNLP 2023. Association for Computational Linguistics, 2023, pp. 14935–14956. https://doi.org/10.18653/v1/2023.findings-emnlp.996.
8. Pourreza, M.; Rafiei, D. DIN-SQL: Decomposed In-Context Learning of Text-to-SQL with Self-Correction. In Proceedings of the Advances in Neural Information Processing Systems 36, 2023.
9. Talaei, S.; Pourreza, M.; Chang, Y.C.; Mirhoseini, A.; Saberi, A. CHESS: Contextual Harnessing for Efficient SQL Synthesis. *arXiv preprint arXiv:2405.16755* **2024**, [2405.16755]. https://doi.org/10.48550/arXiv.2405.16755.
10. Lee, S.; Ko, H. Training a Team of Language Models as Options to Build an SQL-Based Memory. *Applied Sciences* **2025**, *15*, 11399. https://doi.org/10.3390/app152111399.
11. Wang, B.; Ren, C.; Yang, J.; Liang, X.; Bai, J.; Chai, L.; Yan, Z.; Zhang, Q.W.; Yin, D.; Sun, X.; et al. MAC-SQL: A Multi-Agent Collaborative Framework for Text-to-SQL. In Proceedings of the 31st International Conference on Computational Linguistics, 2025, pp. 540–557.
12. Shao, Z.; Cai, S.; Lin, R.; Ming, Z. Enhancing Text-to-SQL with Question Classification and Multi-Agent Collaboration. In Proceedings of the Findings of the Association for Computational Linguistics: NAACL 2025, 2025, pp. 4340–4349. https://doi.org/10.18653/v1/2025.findings-naacl.245.
13. Xia, H.; Jiang, F.; Deng, N.; Wang, C.; Zhao, G.; Mihalcea, R.; Zhang, Y. R^3: "This Is My SQL, Are You with Me?" A Consensus-Based Multi-Agent System for Text-to-SQL Tasks. In Proceedings of the 4th Table Representation Learning Workshop, 2025, pp. 34–46. https://doi.org/10.18653/v1/2025.trl-1.4.
14. Lei, W.; Wang, W.; Ma, Z.; Gan, T.; Lu, W.; Kan, M.Y.; Chua, T.S. Re-examining the Role of Schema Linking in Text-to-SQL. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, 2020, pp. 6943–6954. https://doi.org/10.18653/v1/2020.emnlp-main.564.

15. Liu, A.; Hu, X.; Lin, L.; Wen, L. Semantic Enhanced Text-to-SQL Parsing via Iteratively Learning Schema Linking Graph. In Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. Association for Computing Machinery, 2022, pp. 1021–1030. https://doi.org/10.1145/3534678.3539294.
16. Bozdemir, M.; Bilgin, M. Schema Retrieval with Embeddings and Vector Stores Using Retrieval-Augmented Generation and LLM-Based SQL Query Generation. *Applied Sciences* **2026**, *16*, 586. https://doi.org/10.3390/app16020586.
17. Liu, G.; Tan, Y.; Zhong, R.; Xie, Y.; Zhao, L.; Wang, Q.; Hu, B.; Li, Z. Solid-SQL: Enhanced Schema-Linking-Based In-Context Learning for Robust Text-to-SQL. In Proceedings of the 31st International Conference on Computational Linguistics, 2025, pp. 9793–9803.
18. Hao, Z.; Huang, J.; Shi, S.; Cai, R.; Xu, B. GenLink: Generation-Driven Schema Linking via Multi-Model Learning for Text-to-SQL. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, 2025, pp. 29892–29905. https://doi.org/10.18653/v1/2025.emnlp-main.1518.
19. Wang, Y.; Liu, P.; Yang, X. LinkAlign: Scalable Schema Linking for Real-World Large-Scale Multi-Database Text-to-SQL. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, 2025, pp. 977–991. https://doi.org/10.18653/v1/2025.emnlp-main.51.
20. Safdarian, A.; Mohammadi, M.; Bashirloo, E.J.; Naderi, M.S.; Faili, H. SchemaGraphSQL: Efficient Schema Linking with Pathfinding Graph Algorithms for Text-to-SQL on Large-Scale Databases. In Proceedings of the Findings of the Association for Computational Linguistics: EACL 2026. Association for Computational Linguistics, 2026, pp. 2585–2599. https://doi.org/10.18653/v1/2026.findings-eacl.134.
21. Nahid, M.M.H.; Rafiei, D.; Zhang, W.; Zhang, Y. Rethinking Schema Linking: A Context-Aware Bidirectional Retrieval Approach for Text-to-SQL. In Proceedings of the Findings of the Association for Computational Linguistics: EACL 2026, 2026, pp. 4516–4546. https://doi.org/10.18653/v1/2026.findings-eacl.236.
22. Zhong, R.; Yu, T.; Klein, D. Semantic Evaluation for Text-to-SQL with Distilled Test Suites. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, 2020, pp. 396–411. https://doi.org/10.18653/v1/2020.emnlp-main.29.
23. Sun, R.; Arik, S.O.; Sinha, R.; Nakhost, H.; Dai, H.; Yin, P.; Pfister, T. SQLPrompt: In-Context Text-to-SQL with Minimal Labeled Data. In Proceedings of the Findings of the Association for Computational Linguistics: EMNLP 2023. Association for Computational Linguistics, 2023, pp. 542–550. https://doi.org/10.18653/v1/2023.findings-emnlp.39.
24. Mao, W.; Wang, R.; Guo, J.; Zeng, J.; Gao, C.; Han, P.; Liu, C. Enhancing Text-to-SQL Parsing through Question Rewriting and Execution-Guided Refinement. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2024. Association for Computational Linguistics, 2024, pp. 2009–2024. https://doi.org/10.18653/v1/2024.findings-acl.120.
25. Xie, Y.; Jin, X.; Xie, T.; Lin, M.; Chen, L.; Yu, C.; Cheng, L.; Zhuo, C.; Hu, B.; Li, Z. Decomposition for Enhancing Attention: Improving LLM-Based Text-to-SQL through a Workflow Paradigm. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2024, 2024, pp. 10796–10816. https://doi.org/10.18653/v1/2024.findings-acl.641.
26. Luo, R.; Wang, L.; Lin, B.; Lin, Z.; Yang, Y. PTD-SQL: Partitioning and Targeted Drilling with LLMs in Text-to-SQL. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, 2024, pp. 3767–3799. https://doi.org/10.18653/v1/2024.emnlp-main.221.
27. Zhai, B.; Xu, C.; He, Y.; Yao, Z. Optimizing Reasoning for Text-to-SQL with Execution Feedback. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2025, 2025, pp. 19206–19218. https://doi.org/10.18653/v1/2025.findings-acl.982.
28. Xie, W.; Dai, Y.; Jiang, W. SDE-SQL: Enhancing Text-to-SQL Generation in Large Language Models via Self-Driven Exploration with SQL Probes. In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics, Volume 1: Long Papers. Association for Computational Linguistics, 2026, pp. 2507–2525. https://doi.org/10.18653/v1/2026.acl-long.116.
29. Zhang, J.; Yang, H.; Miao, K.; Zhang, R.; Pi, R.; Gao, J.; Zhou, X. ExeSQL: Self-Taught Text-to-SQL Models with Execution-Driven Bootstrapping for SQL Dialects. In Proceedings of the Findings of the Association for Computational Linguistics: EMNLP 2025, 2025, pp. 24305–24326. https://doi.org/10.18653/v1/2025.findings-emnlp.1320.
30. Barrows, C.; Bloom, A.; Ehlen, A.; Ikäheimo, J.; Jorgenson, J.; Krishnamurthy, D.; Lau, J.; McBennett, B.; O'Connell, M.; Preston, E.; et al. The IEEE Reliability Test System: A Proposed 2019 Update. *IEEE Transactions on Power Systems* **2020**, *35*, 119–127. https://doi.org/10.1109/TPWRS.2019.2925557.
31. Meinecke, S.; Sarajlić, D.; Drauz, S.R.; Klettke, A.; Lauven, L.P.; Rehtanz, C.; Moser, A.; Braun, M. SimBench—A Benchmark Dataset of Electric Power Systems to Compare Innovative Solutions Based on Power Flow Analysis. *Energies* **2020**, *13*, 3290. https://doi.org/10.3390/en13123290.
32. Bian, H.; Li, G.; Wang, Y.; Fu, Q.; Shen, J.; Yao, X.; Wu, Z. DKASQL: Dynamic Knowledge Adaptation for Domain-Specific Text-to-SQL. *Applied Sciences* **2025**, *15*, 11121. https://doi.org/10.3390/app152011121.
33. Li, S.; He, Y.; Ao, L.; Qi, R. RGISQL: Integrating Refined Grammatical Information into Relational Graph Neural Network for Text-to-SQL Task. *Applied Sciences* **2024**, *14*, 10359. https://doi.org/10.3390/app142210359.

34. Zhou, R.; Zhang, F. Refining Zero-Shot Text-to-SQL Benchmarks via Prompt Strategies with Large Language Models. *Applied Sciences* **2025**, *15*, 5306. https://doi.org/10.3390/app15105306.
35. Cameron, A.C.; Gelbach, J.B.; Miller, D.L. Bootstrap-Based Improvements for Inference with Clustered Errors. *The Review of Economics and Statistics* **2008**, *90*, 414–427. https://doi.org/10.1162/rest.90.3.414.
36. Canay, I.A.; Romano, J.P.; Shaikh, A.M. Randomization Tests under an Approximate Symmetry Assumption. *Econometrica* **2017**, *85*, 1013–1030. https://doi.org/10.3982/ECTA13081.
37. Holm, S. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* **1979**, *6*, 65–70.